



VYTAL Project

Submitted by:

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Graduation Project Part-I I

Introduction & Motivation

Athlete health monitoring is essential for improving performance, preventing physiological risks, and supporting timely medical intervention. However, large volumes of physiological data can be difficult to interpret quickly, especially during training and competition.

VYTAL was developed as an AI-powered athlete health monitoring platform that combines physiological analytics, evidence-based risk classification, interactive dashboards, and AI-generated interpretations. The system transforms physiological measurements into actionable insights that support coaches and medical teams in making informed decisions.



Problem Statement

- Physiological measurements are often presented as isolated numerical values, making risk identification difficult.
- Existing monitoring systems typically focus on data collection and visualization while providing limited automated interpretation.
- Coaches and medical teams may require significant effort to manually assess athlete conditions and identify at-risk individuals.
- There is a need for an intelligent platform that combines monitoring, risk classification, visualization, and AI-assisted interpretation within a unified environment.



Project Objectives

- Develop an AI-powered athlete health monitoring platform using Power BI and Python.
- Implement an evidence-based health classification framework using physiological thresholds.
- Provide interactive dashboards for monitoring, trend analysis, and athlete comparison.
- Integrate Gemini LLM to generate automated health interpretations and analytical reports.

Related Work

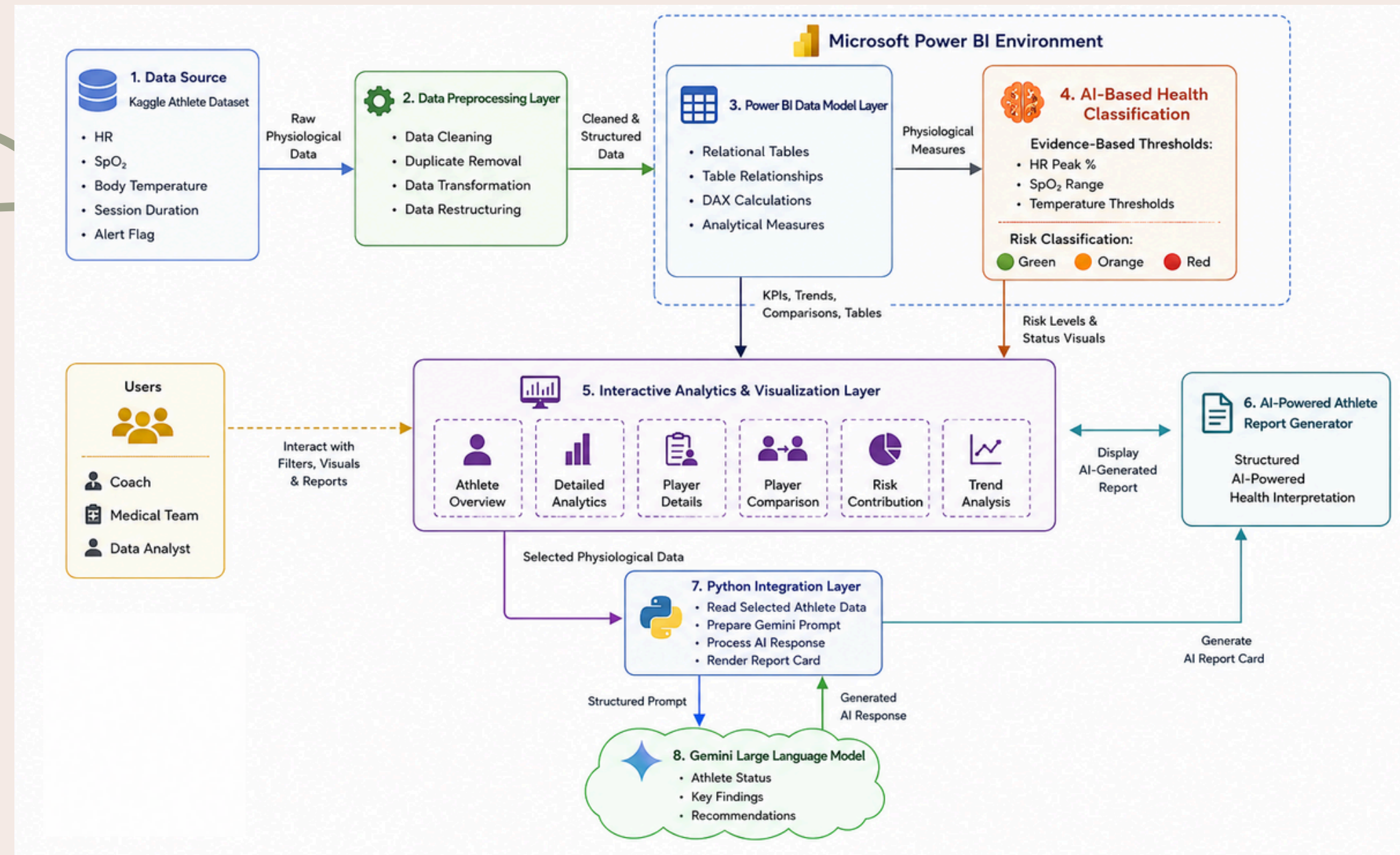
Previous studies have demonstrated the value of dashboard technologies, AI-based decision support systems, and athlete monitoring frameworks. However, most existing solutions focus on individual components rather than a unified intelligent monitoring environment.

Research Gap:

- Monitoring systems often lack automated interpretation.
- Dashboard solutions focus mainly on visualization.
- AI applications are rarely integrated directly with physiological monitoring workflows.

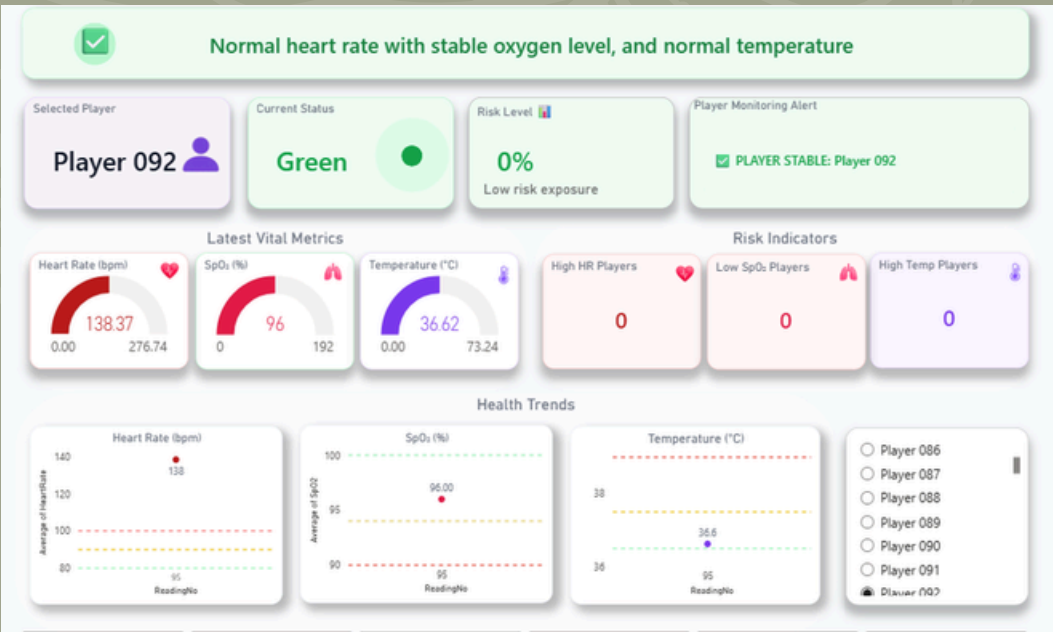
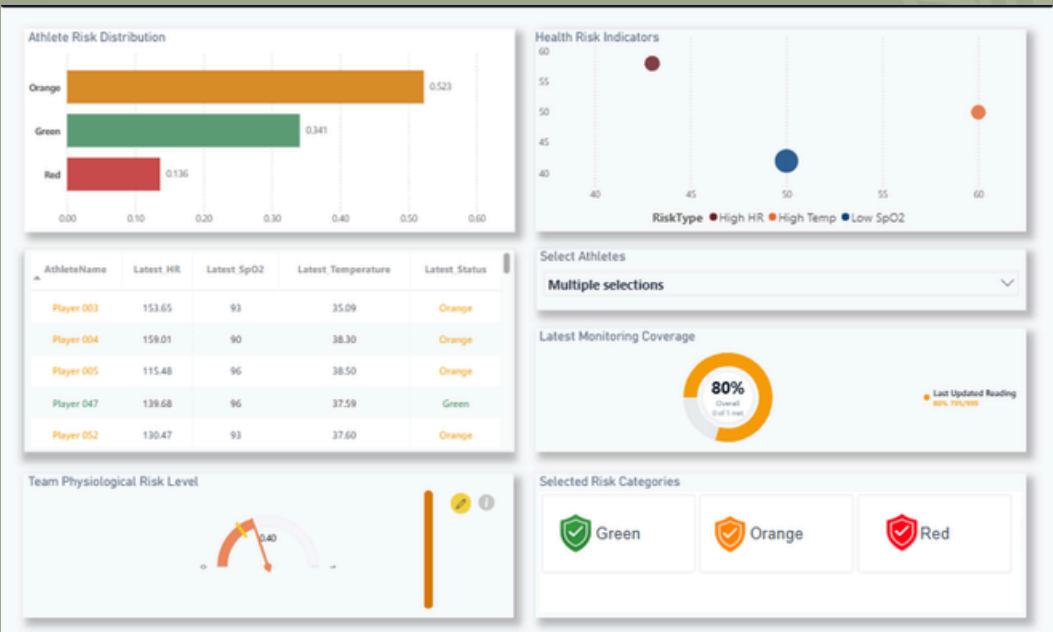
VYTAL addresses this gap by integrating physiological monitoring, evidence-based risk classification, interactive analytics, and AI-assisted interpretation within a single platform.

System Architecture

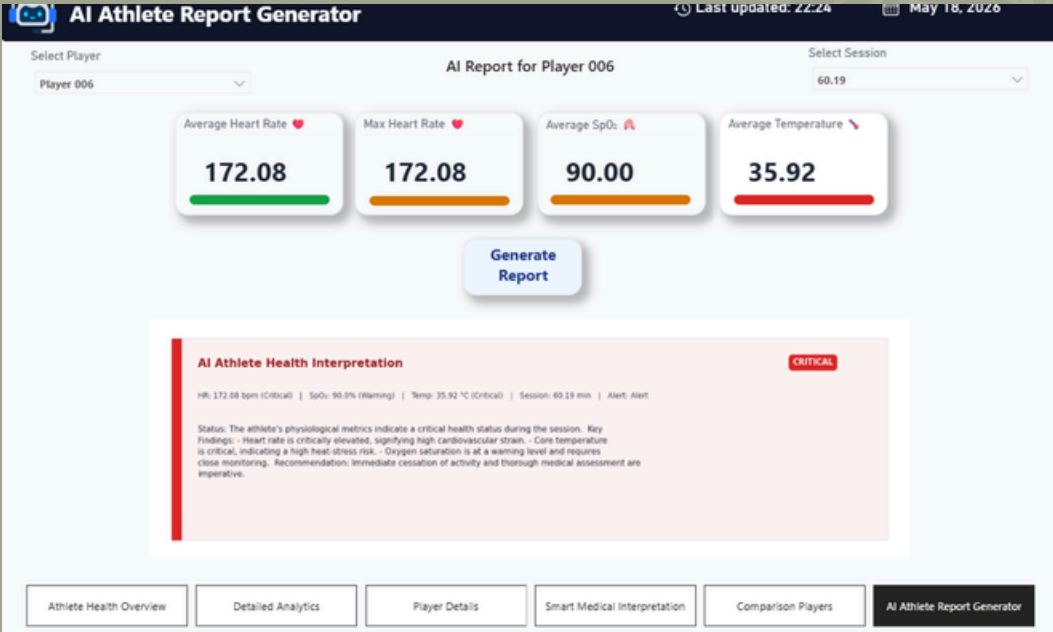
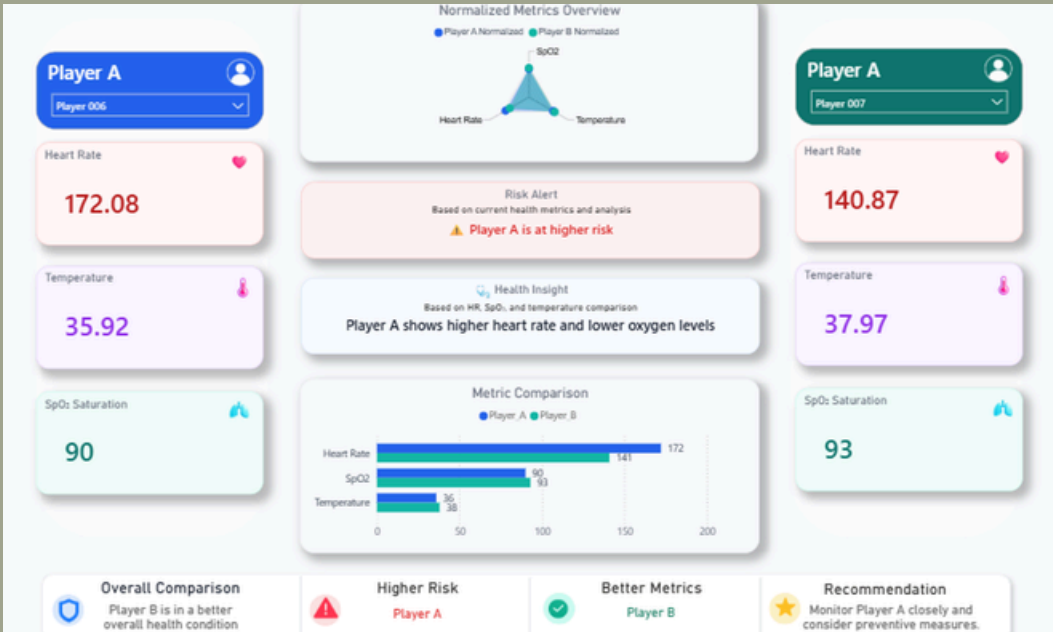
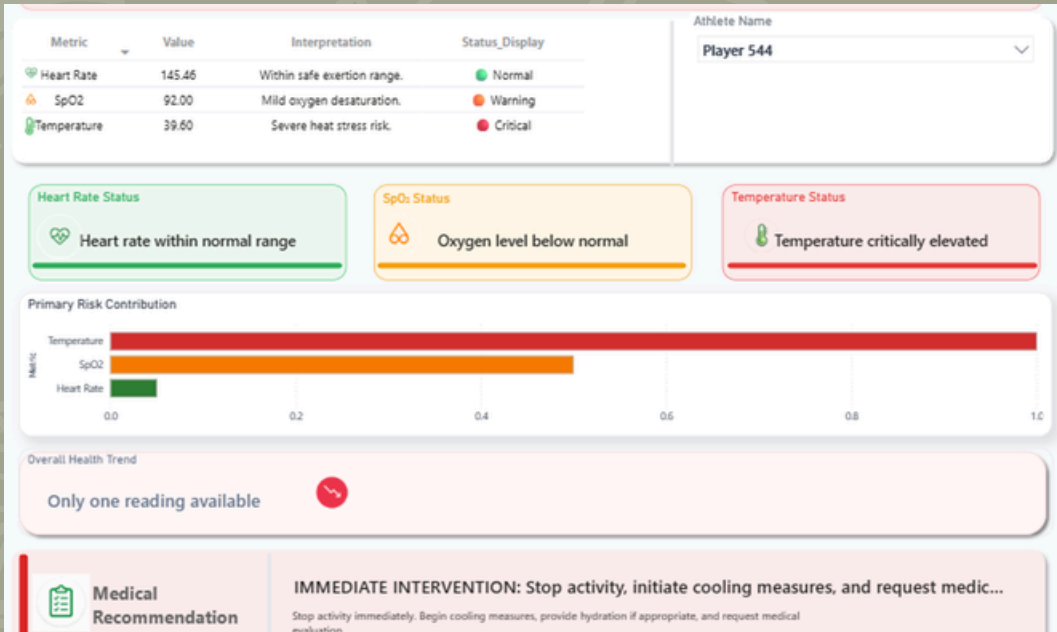


Implementation





Results



Future Work

- Real-time integration with wearable sensors and IoT devices.
- Inclusion of additional physiological indicators.
- Development of predictive analytics and injury-risk forecasting models.
- Mobile application integration and personalized athlete recommendations.

Conclusion

- Successfully integrated physiological monitoring, business intelligence, and artificial intelligence within a unified platform.
- Automated risk classification improved athlete health assessment and monitoring efficiency.
- AI-generated interpretations enhanced understanding of physiological data and supported decision-making.



REFERENCES

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- [2] Ahmadi et al., 2022 – Hospital Dashboard Requirements and Challenges
- [6] Alowais et al., 2024 – Large Language Models in Healthcare
- [10] Dudley et al., 2023 – Athlete Monitoring and Physiological Load Assessment
- [12] Stevens et al., 2023 – Core Body Temperature Monitoring in Sports
- [14] Wang et al., 2024 – AI-Based Clinical Decision Support Systems
- [16] Athletic Training Energy Expenditure Dataset (Kaggle, 2024)
- [17-19] Physiological Threshold References for Heart Rate, SpO₂, and Body Temperature Classification

Live Demonstration

1. Athlete Health Overview Dashboard
2. Detailed Analytics Dashboard
3. Player Details Dashboard
4. Smart Medical Interpretation Module
5. Comparison Players Dashboard
6. AI Athlete Report Generator



Thank You